

Full Model Specification and Reproducibility Notes

This file presents the complete mathematical model to help readers clearly understand its structure and implementation. It includes the notation, objective functions, constraints, and modeling assumptions, together with additional explanations that clarify the modeling choices and improve the model's transparency and reproducibility. Part II describes the fixed input data and parameters prepared before optimization and does not introduce additional decision constraints.

Part I: Complete Mathematical Formulation

Notation

Sets

Symbol	Definition
s	Raw materials suppliers, indexed by $s = 1, \dots, S $.
c	Cast iron manufacturers, indexed by $c = 1, \dots, C $.
p	Iron ingot manufacturers, indexed by $p = 1, \dots, P $.
d^1	Automobile-industry customer demand points, indexed by $d^1 = 1, \dots, D^1 $.
d^2	Kitchen-appliance customer demand points, indexed by $d^2 = 1, \dots, D^2 $.
d^3	Architecture-decoration customer demand points, indexed by $d^3 = 1, \dots, D^3 $.
f	Refurbishing centers, indexed by $f = 1, \dots, F $.
m	Remanufacturing centers, indexed by $m = 1, \dots, M $.
a	Recycle centers, indexed by $a = 1, \dots, A $.
k	Collection centers, indexed by $k = 1, \dots, K $.
w	Disposal centers, indexed by $w = 1, \dots, W $.
t	Time periods, indexed by $t = 1, \dots, T $.
r^1	The single homogeneous iron material, with $R^1 = \{\text{Fe}\}$ and $ R^1 = 1$.
r^2	Other materials, indexed by $r^2 = 1, \dots, R^2 $.
u	Types of products, indexed by $u = 1, \dots, U $.
n	Scenarios, indexed by $n = 1, \dots, N $.
i	IoT systems in facilities, indexed by $i = 1, \dots, I $.

Parameters

Parameter	Definition
λ	Weight assigned to expected scenario cost in the robust economic objective (dimensionless).
ν	Available budget for constructing facilities equipped with IoT systems (MU).
θ	Penalty coefficient for each scenario-level feasibility margin (MU/tonne).
ψ_u	The iron-ingot share required to produce one unit of product type u ; together with the additive-material shares, it closes the product mass balance. <i>Unit: tonne/tonne.</i>
ϕ	The net modeled iron-input requirement per unit of usable ingot output. <i>Unit: tonne/tonne.</i>
δ	The recycling rate achieved at recycling centers. <i>Unit: tonne/tonne.</i>
γ	The remanufacturing rate for producing new products by architectural decoration manufacturers at remanufacturing centers. <i>Unit: tonne/tonne.</i>
ω	The refurbishment rate for repairing defective products for delivery to customers at refurbishing centers. <i>Unit: tonne/tonne.</i>

Parameter	Definition
β	The return rate of products used by customers. <i>Unit: tonne/tonne.</i>
$\varphi_{r^2,u}$	Additive-material mass fraction of material r^2 in product type u . <i>Unit: tonne/tonne.</i>
$\mu_{k,t}^\sigma$	The rate that is determined at the collection centers k to send the products to σ ($\sigma = 1$ (recycle), 2 (refurbishing), 3 (remanufacturing), 4 (disposal)) centers at period t . <i>Unit: dimensionless.</i>
$c_{r^1,s,p,n}$	Purchasing cost per ton iron r^1 from raw materials suppliers s to iron ingot manufacturers p under scenario n . <i>Unit: MU/tonne.</i>
$c_{r^2,s,c,n}$	Purchasing cost per ton other materials r^2 from raw materials suppliers s to cast iron manufacturers c under scenario n . <i>Unit: MU/tonne.</i>
p_σ	Production cost per ton for σ ($\sigma \in \{p, c, m\}$). <i>Unit: MU/tonne of usable output.</i>
o_k	Recovery-oriented sorting, grading, consolidation, and preparation cost per tonne routed from collection center k to recycling, refurbishing, or remanufacturing. <i>Unit: MU/tonne.</i>
o_σ	Operating cost per tonne of usable recovered output for σ ($\sigma \in \{f, m, a\}$). <i>Unit: MU/tonne.</i>
o_w	Receiving, basic handling, and disposal operating cost per tonne received at disposal facility w . <i>Unit: MU/tonne.</i>
cap_π	Production capacity of π ($\pi \in \{c, p, f, m, a, w\}$). <i>Unit: tonne/period of usable output or throughput.</i>
cap_{s,r^1}	Supplying capacity of supply iron r^1 for raw materials suppliers s . <i>Unit: tonne/period.</i>
cap_{s,r^2}	Supplying capacity of supply other material r^2 for raw materials suppliers s . <i>Unit: tonne/period.</i>
$f_{\sigma,i}$	Fixed cost of construction σ ($\sigma \in \{c, p, f, m, a, w\}$) based on IoT systems i . <i>Unit: MU.</i>
$t_{r^1,s,p,n}$	Transportation cost per tonne of iron r^1 from raw materials suppliers s to iron ingot manufacturers p under scenario n . <i>Unit: MU/tonne.</i>
$t_{r^2,s,c,n}$	Transportation cost per tonne of other material r^2 from raw materials suppliers s to cast iron manufacturers c under scenario n . <i>Unit: MU/tonne.</i>
$t_{p,c}$	Transportation cost from iron ingot manufacturers p to cast iron manufacturers c . <i>Unit: MU/tonne.</i>
t_{c,d^1}	Transportation cost from cast iron manufacturers c to automobile industry manufacturers d^1 . <i>Unit: MU/tonne.</i>
t_{c,d^2}	Transportation cost from cast iron manufacturers c to kitchen appliance manufacturers d^2 . <i>Unit: MU/tonne.</i>
$t_{k,f}$	Transportation cost from collection centers k to refurbishing centers f . <i>Unit: MU/tonne.</i>
$t_{k,m}$	Transportation cost from collection centers k to remanufacturing centers m . <i>Unit: MU/tonne.</i>
$t_{k,a}$	Transportation cost from collection centers k to recycle centers a . <i>Unit: MU/tonne.</i>
$t_{k,w}$	Transportation cost from collection centers k to disposal centers w . <i>Unit: MU/tonne.</i>
$t_{a,p}$	Transportation cost from recycle centers a to iron ingot manufacturers p . <i>Unit: MU/tonne.</i>
t_{f,d^1}	Transportation cost from refurbishing centers f to automobile industry manufacturers d^1 . <i>Unit: MU/tonne.</i>
t_{f,d^2}	Transportation cost from refurbishing centers f to kitchen appliance manufacturers d^2 . <i>Unit: MU/tonne.</i>
t_{m,d^3}	Transportation cost from remanufacturing centers m to architecture decoration manufacturers d^3 . <i>Unit: MU/tonne.</i>
$e_{c,t}^q$	Operating cost of utility q at cast-iron manufacturer c in period t , where $q = 1$ denotes electricity and $q = 2$ denotes water. The aggregate coefficient used in the computational implementation is $e_{c,t} = e_{c,t}^1 + e_{c,t}^2$. <i>Unit: MU/active plant-period.</i>

Parameter	Definition
$d_{\sigma,t,n}$	Aggregate demand over product types for σ ($\sigma \in \{d^1, d^2, d^3\}$) at period t under scenario n . <i>Unit: tonne/period.</i>
g_σ	GHG emissions associated with one tonne of material processed or usable output provided at facility σ ($\sigma \in \{c, p, s, f, m, a, w\}$). <i>Unit: kg CO₂-eq/tonne.</i>
$j_{\sigma,i}$	Total job creation in σ ($\sigma \in \{f, m, a, p, c, w\}$), for IoT system i . <i>Unit: jobs.</i>
w_c^σ	Consumption per active period of resource type σ ($\sigma = 1$ for electricity and $\sigma = 2$ for water) at cast iron manufacturers c , where electricity consumption is measured in kJ and water consumption is measured in liters. These per-period values are constant across periods and scenarios in the present formulation.
π_n	Scenario probability used only in the robust economic objective. <i>Unit: dimensionless.</i>
$s_{r^1,p,t}$	Safety stock of iron r^1 in iron ingot manufacturers p at period t . <i>Unit: tonne.</i>
$s_{r^2,c,t}$	Safety stock of other materials r^2 in cast iron manufacturers c at period t . <i>Unit: tonne.</i>
$h_{\sigma,i}$	Time of worker's training in σ ($\sigma \in \{f, m, a, p, c, w\}$), for IoT system i . <i>Unit: worker-hour.</i>
h_t	Total time of worker's training in period t . <i>Unit: worker-hour.</i>
I_r	Inventory cost of r ($r \in \{r^1, r^2\}$) stock. <i>Unit: MU/(tonne-period).</i>
κ_c^E	Location-specific carbon-equivalent conversion factor for electricity consumed at cast-iron manufacturer c , measured in kg CO ₂ -eq per kJ.
κ^W	Baseline carbon-equivalent conversion factor for water consumption, measured in kg CO ₂ -eq per liter of water consumed. In the case study it is obtained by converting the mean value reported in the cited potable-water life-cycle meta-analysis; it is therefore a proxy rather than a universal or site-specific factor.
$\{M_\ell\}_{\ell \in \{p,c\}}$	Data-derived Big- M upper bounds used only to link positive flows to the activation of the corresponding candidate facility. <i>Unit: tonne/period.</i>
$\bar{I}_{r^1,p,t}, \bar{I}_{r^2,c,t}$	Valid cumulative-availability upper bounds for end-of-period iron and additive inventories. Their numerical values are computed before model generation using the relations in Part II. <i>Unit: tonne.</i>

All material flows, inventories, demands, safety stocks, and the shared feasibility margins are measured in tonnes; per-period supply and processing capacities are measured in tonnes per period. MU denotes the monetary unit used consistently by the dataset. Unit purchasing, production, operation, transportation, and penalty coefficients are in MU per tonne; fixed costs are in MU; $e_{c,t}^q$ is in MU per active plant-period; and inventory holding cost is in MU per tonne-period. Process-emission factors are in kg CO₂-eq per tonne, w_c^1 is in kJ per period, w_c^2 is in litres per period, κ_c^E is in kg CO₂-eq/kJ, and κ^W is in kg CO₂-eq/L. Job coefficients are in jobs and training parameters are in worker-hours. All rates, allocation shares, probabilities, and λ are dimensionless.

Decision variables

Variable	Definition
$X_{r^1,s,p,t,n}$	Iron shipped from supplier s to ingot plant p .
$X_{r^2,s,c,t,n}$	Additive material r^2 shipped from supplier s to cast-iron plant c .
$X_{p,c,t,n}$	Usable ingot shipped from ingot plant p to cast-iron plant c .
$X_{u,c,d^1,t,n}, X_{u,c,d^2,t,n}$	Fresh product u shipped from cast-iron plant c to demand groups d^1 and d^2 .
$X_{d^1,k,t,n}, X_{d^2,k,t,n}$	Returned products shipped from demand groups d^1 and d^2 to collection centre k .
$X_{k,a,t,n}, X_{k,f,t,n}, X_{k,m,t,n}, X_{k,w,t,n}$	Aggregated returns shipped from collection centre k to recycling, refurbishing, remanufacturing, and disposal facilities.
$X_{r^1,a,p,t,n}$	Recycled iron shipped from recycling centre a to ingot plant p .
$X_{u,f,d^1,t,n}, X_{u,f,d^2,t,n}$	Refurbished product u shipped to demand groups d^1 and d^2 .

Variable	Definition
$X_{u,m,d^3,t,n}$	Remanufactured product u shipped to architecture/decor demand point d^3 .
$IN_{r^1,p,t,n}, IN_{r^2,c,t,n}$	End-of-period inventories of iron and additive materials.
$\varepsilon_n^1, \varepsilon_n^2, \varepsilon_n^3$	Customer-group-specific, scenario-level shared feasibility margins.
w_n	Scenario cost.
φ_n^+, φ_n^-	Positive and negative absolute-deviation components of scenario cost.
$Y_{\ell,i}$	Binary variable equal to one if candidate facility $\ell \in \{p, c, f, m, a, w\}$ is activated with IoT alternative i .

Objective functions:

Economic objective:

$$\begin{aligned} \min Z_{RSP}^{Cost} = & \left[\lambda \left(\sum_n \pi_n w_n \right) \right. \\ & + \left((1 - \lambda) \times \sum_n \pi_n [\varphi_n^+ + \varphi_n^-] \right) \\ & \left. + \left(\theta \times \sum_n \pi_n [\varepsilon_n^1 + \varepsilon_n^2 + \varepsilon_n^3] \right) \right] \end{aligned} \quad (1)$$

The first term minimizes probability-weighted scenario cost. The second penalizes the absolute deviation of each scenario cost from the probability-weighted mean and therefore discourages solutions with highly uneven economic performance across scenarios. The third penalizes demand infeasibility. Each ε_n^j is shared by all demand-point and period constraints of customer group j in scenario n ; it is therefore a scenario-level feasibility margin, not the sum of node-period shortages. This is why ε_n^j has only the scenario index in (1).

Full scenario-cost expression:

$$\begin{aligned} w_n = & \sum_{r^1,s,p,t} c_{r^1,s,p,n} X_{r^1,s,p,t,n} + \sum_{r^2,s,c,t} c_{r^2,s,c,n} X_{r^2,s,c,t,n} \\ & + \sum_{p,c,t} p_p X_{p,c,t,n} + \sum_c p_c \left(\sum_{u,d^1,t} X_{u,c,d^1,t,n} + \sum_{u,d^2,t} X_{u,c,d^2,t,n} \right) \\ & + \sum_{u,m,d^3,t} p_m X_{u,m,d^3,t,n} \\ & + \sum_{k,t} o_k \left(\sum_a X_{k,a,t,n} + \sum_f X_{k,f,t,n} + \sum_m X_{k,m,t,n} \right) \\ & + \sum_f o_f \left(\sum_{u,d^1,t} X_{u,f,d^1,t,n} + \sum_{u,d^2,t} X_{u,f,d^2,t,n} \right) \\ & + \sum_{u,m,d^3,t} o_m X_{u,m,d^3,t,n} + \sum_{r^1,a,p,t} o_a X_{r^1,a,p,t,n} + \sum_{k,w,t} o_w X_{k,w,t,n} \\ & + \sum_{r^1,s,p,t} t_{r^1,s,p,n} X_{r^1,s,p,t,n} + \sum_{r^2,s,c,t} t_{r^2,s,c,n} X_{r^2,s,c,t,n} \\ & + \sum_{p,c,t} t_{p,c} X_{p,c,t,n} + \sum_{u,c,d^1,t} t_{c,d^1} X_{u,c,d^1,t,n} \\ & + \sum_{u,c,d^2,t} t_{c,d^2} X_{u,c,d^2,t,n} + \sum_{k,f,t} t_{k,f} X_{k,f,t,n} \end{aligned}$$

$$\begin{aligned}
& + \sum_{k,m,t} t_{k,m} X_{k,m,t,n} + \sum_{k,a,t} t_{k,a} X_{k,a,t,n} + \sum_{k,w,t} t_{k,w} X_{k,w,t,n} \\
& + \sum_{r^1,a,p,t} t_{a,p} X_{r^1,a,p,t,n} + \sum_{u,f,d^1,t} t_{f,d^1} X_{u,f,d^1,t,n} \\
& + \sum_{u,f,d^2,t} t_{f,d^2} X_{u,f,d^2,t,n} + \sum_{u,m,d^3,t} t_{m,d^3} X_{u,m,d^3,t,n} \\
& + \sum_{p,i} f_{p,i} Y_{p,i} + \sum_{c,i} f_{c,i} Y_{c,i} + \sum_{f,i} f_{f,i} Y_{f,i} + \sum_{m,i} f_{m,i} Y_{m,i} \\
& + \sum_{a,i} f_{a,i} Y_{a,i} + \sum_{w,i} f_{w,i} Y_{w,i} + \sum_{c,i,t} (e_{c,t}^1 + e_{c,t}^2) Y_{c,i} \\
& + \sum_{r^1,p,t} I_{r^1} I N_{r^1,p,t,n} + \sum_{r^2,c,t} I_{r^2} I N_{r^2,c,t,n}, \quad \forall n.
\end{aligned} \tag{2}$$

Robust cost-deviation relation:

$$w_n - \sum_{n' \in N} \pi_{n'} w_{n'} = \varphi_n^+ - \varphi_n^- \quad \forall n \tag{3}$$

Environmental objective function:

$$\begin{aligned}
\min Z^{environmental} = & \left[\sum_c g_c \left(\sum_{u,d^1,t,n} X_{u,c,d^1,t,n} + \sum_{u,d^2,t,n} X_{u,c,d^2,t,n} \right) \right. \\
& + \sum_{p,c,t,n} g_p X_{p,c,t,n} + \sum_{r^1,s,p,t,n} g_s X_{r^1,s,p,t,n} \\
& + \sum_{r^2,s,c,t,n} g_s X_{r^2,s,c,t,n} \\
& + \sum_f g_f \left(\sum_{u,d^1,t,n} X_{u,f,d^1,t,n} + \sum_{u,d^2,t,n} X_{u,f,d^2,t,n} \right) \\
& + \sum_{u,m,d^3,t,n} g_m X_{u,m,d^3,t,n} + \sum_{r^1,a,p,t,n} g_a X_{r^1,a,p,t,n} \\
& + \sum_{k,w,t,n} g_w X_{k,w,t,n} \\
& \left. + |N| \sum_{c,i} Y_{c,i} \left(\sum_t \kappa_c^E w_c^1 + \sum_t \kappa_c^W w_c^2 \right) \right] \tag{4}
\end{aligned}$$

$$\max Z^{social} = \left[\sum_{p,i} j_{p,i} Y_{p,i} + \sum_{c,i} j_{c,i} Y_{c,i} + \sum_{f,i} j_{f,i} Y_{f,i} + \sum_{m,i} j_{m,i} Y_{m,i} + \sum_{a,i} j_{a,i} Y_{a,i} + \sum_{w,i} j_{w,i} Y_{w,i} \right] \tag{5}$$

The social objective counts the jobs associated with the selected facility–IoT combinations. Because $Y_{\ell,i}$ is a strategic, scenario-independent decision, job creation is counted once and is not summed over scenarios or periods. When (5) is optimized by itself, operational flows may be non-unique because the objective depends only on Y ; the reported Pareto solutions should therefore be obtained within the stated multiobjective framework, where the economic and environmental criteria also control flows.

Constraints:

Supplier production raw materials capacity constraint:

$$\sum_p X_{r^1,s,p,t,n} \leq cap_{s,r^1} \quad \forall r^1, s, t, n \tag{6}$$

$$\sum_c X_{r^2,s,c,t,n} \leq cap_{s,r^2} \quad \forall r^2, s, t, n \tag{7}$$

$$\sum_{u,d^1} X_{u,c,d^1,t,n} + \sum_{u,d^2} X_{u,c,d^2,t,n} \leq cap_c \sum_i Y_{c,i} \quad \forall c, t, n \quad (8)$$

$$\sum_c X_{p,c,t,n} \leq cap_p \sum_i Y_{p,i} \quad \forall p, t, n \quad (9)$$

$$\sum_{u,d^1} X_{u,f,d^1,t,n} + \sum_{u,d^2} X_{u,f,d^2,t,n} \leq cap_f \sum_i Y_{f,i} \quad \forall f, t, n \quad (10)$$

$$\sum_{u,d^3} X_{u,m,d^3,t,n} \leq cap_m \sum_i Y_{m,i} \quad \forall m, t, n \quad (11)$$

$$\sum_{r^1,p} X_{r^1,a,p,t,n} \leq cap_a \sum_i Y_{a,i} \quad \forall a, t, n \quad (12)$$

$$\sum_k X_{k,w,t,n} \leq cap_w \sum_i Y_{w,i} \quad \forall w, t, n \quad (13)$$

Constraints (6)–(7) restrict each supplier's shipments of each material to its available per-period supply. Constraints (8)–(13) impose the relevant usable-output or throughput capacity and simultaneously deactivate output when no IoT alternative is selected. The cost and emission coefficients for production and recovery are applied to these usable outputs, consistently with Assumptions 7 and 8.

The model considers three predefined IoT system alternatives for each candidate facility. Since these alternatives represent mutually exclusive technology choices, a facility can be equipped with at most one IoT system. Therefore, the corresponding binary selection variables are restricted such that the sum of the IoT-selection variables for each facility cannot exceed one. This condition also allows a candidate facility to remain unselected when none of the IoT alternatives is adopted.

Accordingly, $\sum_i Y_{\ell,i}$ can only be either 0 or 1: zero indicates that candidate facility ℓ is inactive, whereas one indicates that it is activated with exactly one of the available IoT systems.

IoT selection and inbound-flow activation constraints:

$$\sum_{i \in I} Y_{\sigma,i} \leq 1 \quad \forall \sigma \in \{f, m, a, p, c, w\}. \quad (14)$$

Equation (14) is the compact form of the six facility-specific IoT-selection conditions. The GAMS implementation expands this expression over the six facility domains because GAMS stores their binary variables separately.

Big- M inbound-flow activation constraints:

$$\sum_{r^1,s} X_{r^1,s,p,t,n} + \sum_{r^1,a} X_{r^1,a,p,t,n} \leq M_p \sum_i Y_{p,i} \quad \forall p, t, n \quad (15)$$

$$\sum_{r^2,s} X_{r^2,s,c,t,n} + \sum_p X_{p,c,t,n} \leq M_c \sum_i Y_{c,i} \quad \forall c, t, n \quad (16)$$

Demand of customers:

Demand is aggregate over product types. Therefore, the sums over u in constraints (17)–(21) allow the available product types to jointly satisfy the requirement of a customer group; no product-specific demand parameter is imposed. The same scenario-level margin ε_n^j appears in every constraint of customer group j ; it does not acquire demand-point or period indices. The architecture/decor constraint applies only for $t > 1$, while $d_{d^3,1,n} = 0$ is fixed in the input data, as documented in Part II.

$$\sum_{u,c} X_{u,c,d^1,t,n} + \varepsilon_n^1 \geq d_{d^1,t,n} \quad \forall d^1, t = 1, n \quad (17)$$

$$\sum_{u,c} X_{u,c,d^1,t,n} + \sum_{u,f} X_{u,f,d^1,t,n} + \varepsilon_n^1 \geq d_{d^1,t,n} \quad \forall d^1, t > 1, n \quad (18)$$

$$\sum_{u,c} X_{u,c,d^2,t,n} + \varepsilon_n^2 \geq d_{d^2,t,n} \quad \forall d^2, t = 1, n \quad (19)$$

$$\sum_{u,c} X_{u,c,d^2,t,n} + \sum_{u,f} X_{u,f,d^2,t,n} + \varepsilon_n^2 \geq d_{d^2,t,n} \quad \forall d^2, t > 1, n \quad (20)$$

$$\sum_{u,m} X_{u,m,d^3,t,n} + \varepsilon_n^3 \geq d_{d^3,t,n} \quad \forall d^3, t > 1, n \quad (21)$$

$$\sum_k X_{d^1,k,t,n} = \beta \times d_{d^1,t,n} \quad \forall d^1, t > 1, n \quad (22)$$

$$\sum_k X_{d^2,k,t,n} = \beta \times d_{d^2,t,n} \quad \forall d^2, t > 1, n \quad (23)$$

Period one is the initialization period. The customer-to-collection variables are fixed to zero in the implementation before model generation rather than introduced as two additional structural equations. Together with nonnegativity and the route-allocation and recovery balances, this setting also fixes every first-period reverse and recovered-output flow to zero. For $t > 1$, the preceding return equations use the exogenous installed-base proxy βd and are not reduced by current-period fulfilled sales.

$$\sum_s X_{r^1,s,p,t,n} = IN_{r^1,p,t,n} + \phi \times \sum_c X_{p,c,t,n} \quad \forall r^1, p, t = 1, n \quad (24)$$

$$\sum_s X_{r^1,s,p,t,n} + \sum_a X_{r^1,a,p,t,n} + IN_{r^1,p,t-1,n} = IN_{r^1,p,t,n} + \phi \times \sum_c X_{p,c,t,n} \quad \forall r^1, p, t > 1, n \quad (25)$$

Both iron balances are imposed for the singleton set $r^1 \in R^1 = \{\text{Fe}\}$. Thus $\phi \sum_c X_{p,c,t,n}$ is the total modeled iron requirement, with no duplication across grades.

Material and collection:

$$\sum_s X_{r^2,s,c,t,n} = IN_{r^2,c,t,n} + \sum_u \varphi_{r^2,u} \left(\sum_{d^1} X_{u,c,d^1,t,n} + \sum_{d^2} X_{u,c,d^2,t,n} \right) \quad \forall r^2, c, t = 1, n \quad (26)$$

$$\sum_s X_{r^2,s,c,t,n} + IN_{r^2,c,t-1,n} = IN_{r^2,c,t,n} + \sum_u \varphi_{r^2,u} \left(\sum_{d^1} X_{u,c,d^1,t,n} + \sum_{d^2} X_{u,c,d^2,t,n} \right) \quad \forall r^2, c, t > 1, n \quad (27)$$

$$\sum_p X_{p,c,t,n} = \sum_u \psi_u \left(\sum_{d^1} X_{u,c,d^1,t,n} + \sum_{d^2} X_{u,c,d^2,t,n} \right) \quad \forall c, t, n \quad (28)$$

$$\sum_a X_{k,a,t,n} = \mu_{k,t}^1 \left(\sum_{d^1} X_{d^1,k,t,n} + \sum_{d^2} X_{d^2,k,t,n} \right) \quad \forall k, t, n \quad (29)$$

$$\sum_f X_{k,f,t,n} = \mu_{k,t}^2 \left(\sum_{d^1} X_{d^1,k,t,n} + \sum_{d^2} X_{d^2,k,t,n} \right) \quad \forall k, t, n \quad (30)$$

$$\sum_m X_{k,m,t,n} = \mu_{k,t}^3 \left(\sum_{d^1} X_{d^1,k,t,n} + \sum_{d^2} X_{d^2,k,t,n} \right) \quad \forall k, t, n \quad (31)$$

$$\sum_w X_{k,w,t,n} = \mu_{k,t}^4 \left(\sum_{d^1} X_{d^1,k,t,n} + \sum_{d^2} X_{d^2,k,t,n} \right) \quad \forall k, t, n \quad (32)$$

Constraints (26)–(27) conserve every additive material over time. Constraint (28) supplies the ingot share required by each product. Product composition shares are validated in the input data to sum to one. Constraints (29)–(32) apply the four predefined route shares; because those shares are validated to sum to one, total collection conservation follows automatically.

Recovery balances and resource constraints:

$$\sum_k (\delta \times X_{k,a,t,n}) = \sum_{r^1,p} X_{r^1,a,p,t,n} \quad \forall a, t, n \quad (33)$$

$$\sum_k (\omega \times X_{k,f,t,n}) = \sum_{u,d^1} X_{u,f,d^1,t,n} + \sum_{u,d^2} X_{u,f,d^2,t,n} \quad \forall f, t, n \quad (34)$$

$$\sum_k (\gamma \times X_{k,m,t,n}) = \sum_{u,d^3} X_{u,m,d^3,t,n} \quad \forall m, t, n \quad (35)$$

Each equality converts recovery-facility input into usable output through its corresponding strictly positive yield. The unrecovered fraction is a process loss outside the usable-output boundary. Because $0 < \delta, \omega, \gamma \leq 1$, the recovery balances and output-capacity constraints also prevent a closed recovery facility from receiving a positive flow; separate Big- M constraints are therefore unnecessary for recycling, refurbishing, and remanufacturing facilities.

$$\sum_{p,i} f_{p,i} Y_{p,i} + \sum_{c,i} f_{c,i} Y_{c,i} + \sum_{f,i} f_{f,i} Y_{f,i} + \sum_{m,i} f_{m,i} Y_{m,i} + \sum_{a,i} f_{a,i} Y_{a,i} + \sum_{w,i} f_{w,i} Y_{w,i} \leq \nu \quad (36)$$

The one-time fixed costs of all selected facility-IoT combinations must fit within the available strategic investment budget ν .

$$s_{r^1,p,t} \sum_i Y_{p,i} \leq IN_{r^1,p,t,n} \leq \bar{I}_{r^1,p,t} \sum_i Y_{p,i} \quad \forall r^1, p, t, n \quad (37)$$

$$s_{r^2,c,t} \sum_i Y_{c,i} \leq IN_{r^2,c,t,n} \leq \bar{I}_{r^2,c,t} \sum_i Y_{c,i} \quad \forall r^2, c, t, n \quad (38)$$

The lower terms impose the required safety stocks. The upper terms use valid cumulative-availability bounds computed before model generation; they do not represent independently estimated warehouse capacities. When a plant is inactive, both inventory and inbound material are forced to zero. The calculation and validation of $\bar{I}$ are reported in Part II.

$$\sum_{w,i} h_{w,i} Y_{w,i} + \sum_{f,i} h_{f,i} Y_{f,i} + \sum_{m,i} h_{m,i} Y_{m,i} + \sum_{a,i} h_{a,i} Y_{a,i} + \sum_{p,i} h_{p,i} Y_{p,i} + \sum_{c,i} h_{c,i} Y_{c,i} \leq h_t \quad \forall t \quad (39)$$

For every period, the training requirement of the selected IoT configuration cannot exceed the available worker-training time h_t . Because the selection variables are strategic and have no period index, this is a conservative availability check rather than a separate training-scheduling model.

$$\begin{aligned} &X_{r^1,s,p,t,n}, X_{r^2,s,c,t,n}, X_{p,c,t,n}, X_{u,c,d^1,t,n}, X_{u,c,d^2,t,n}, X_{u,m,d^3,t,n}, X_{k,a,t,n}, X_{k,f,t,n}, X_{k,m,t,n}, \\ &X_{u,f,d^1,t,n}, X_{u,f,d^2,t,n}, X_{r^1,a,p,t,n}, X_{k,w,t,n}, X_{d^1,k,t,n}, X_{d^2,k,t,n}, \varphi_n^+, \varphi_n^-, \\ &\varepsilon_n^1, \varepsilon_n^2, \varepsilon_n^3, IN_{r^1,p,t,n}, IN_{r^2,c,t,n} \geq 0 \end{aligned} \quad (40)$$

$$Y_{p,i}, Y_{c,i}, Y_{f,i}, Y_{m,i}, Y_{a,i}, Y_{w,i} \in \{0, 1\} \quad (41)$$

Constraint (40) enforces nonnegativity of every physical flow, inventory, feasibility margin, and deviation variable. Constraint (41) makes each IoT selection binary; together with (14), each candidate facility is either closed or activated with exactly one selected alternative.

Part II: Input-Data Preparation and Derived Parameters

Part II records the data rules used by the public implementation. The symbols in this part are fixed inputs or parameters calculated from fixed inputs. They are evaluated before the solver model is generated. Once calculated, the resulting numerical arrays are passed to the relevant equations in Part I.

Data admissibility rules:

The implementation checks the following conditions immediately after loading the raw data:

$$0 \leq \lambda \leq 1, \quad \pi_n > 0 \quad \forall n, \quad \sum_{n \in N} \pi_n = 1, \quad (\text{P1})$$

$$0 < \beta, \delta, \omega, \gamma \leq 1, \quad \phi = 0.20, \quad (\text{P2})$$

$$0 \leq \mu_{k,t}^\sigma \leq 1 \quad \forall k, t, \quad \sigma \in \{1, 2, 3, 4\}, \quad (\text{P3})$$

$$\sum_{\sigma=1}^4 \mu_{k,t}^\sigma = 1 \quad \forall k, t, \quad (\text{P4})$$

$$\psi_u + \sum_{r^2 \in R^2} \varphi_{r^2,u} = 1, \quad \psi_u > 0, \quad \forall u, \quad (\text{P5})$$

$$\varphi_{r^2,u} \geq 0 \quad \forall r^2, u, \quad (\text{P5})$$

$$d_{d^3,1,n} = 0 \quad \forall d^3, n, \quad (\text{P6})$$

$$s_{r^1,p,t} \geq 0, \quad s_{r^2,c,t} \geq 0 \quad \forall r^1, r^2, p, c, t. \quad (\text{P7})$$

Demand, capacity, cost, environmental-impact, budget, and training inputs are also required to be finite and nonnegative. A failed check stops model generation, so an invalid dataset cannot be passed to the optimizer. For the single-scenario computational file, $N = \{n1\}$ and therefore $\pi_{n1} = 1$.

The following relations convert the validated raw data into numerical bounds. They contain no decision variables and are evaluated only once, before the equations of the optimization model are instantiated:

$$S_{r^1}^{cap} = \sum_s cap_{s,r^1}, \quad S_{r^2}^{cap} = \sum_s cap_{s,r^2}, \quad (\text{P8})$$

$$R_{t,n}^{ret} = \begin{cases} 0, & t = 1, \\ \beta \left(\sum_{d^1} d_{d^1,t,n} + \sum_{d^2} d_{d^2,t,n} \right), & t > 1, \end{cases} \quad (\text{P9})$$

$$B_{t,n}^\sigma = \left(\max_k \mu_{k,t}^\sigma \right) R_{t,n}^{ret}, \quad \bar{B}^\sigma = \max_{t,n} B_{t,n}^\sigma, \quad \sigma \in \{1, 2, 3, 4\}, \quad (\text{P10})$$

$$\bar{I}_{r^1,p,t} = tS_{r^1}^{cap} + \delta \max_{n \in N} \sum_{\tau=1}^t B_{\tau,n}^1, \quad \forall r^1 \in \{\text{Fe}\}, p, t, \quad (\text{P11})$$

$$\bar{I}_{r^2,c,t} = tS_{r^2}^{cap}, \quad \forall r^2, c, t, \quad (\text{P12})$$

$$M_p = \min \left\{ S_{\text{Fe}}^{cap} + \delta \bar{B}^1, \phi cap_p + \max_t \bar{I}_{\text{Fe},p,t} \right\}, \quad \forall p, \quad (\text{P13})$$

$$M_c = \min \left\{ \sum_p cap_p + \sum_{r^2} S_{r^2}^{cap}, cap_c + \max_t \sum_{r^2} \bar{I}_{r^2,c,t} \right\}, \quad \forall c, \quad (\text{P14})$$

$$s_{r^1,p,t} \leq \bar{I}_{r^1,p,t}, \quad s_{r^2,c,t} \leq \bar{I}_{r^2,c,t}. \quad (\text{P15})$$

After these calculations, $\bar{I}_{r^1,p,t}$ and $\bar{I}_{r^2,c,t}$ are inserted as fixed right-hand-side coefficients in the inventory bounds of (37)–(38). Likewise, M_p and M_c are inserted in the inbound-flow activation relations (15)–(16). Thus, the formulas above generate parameters; the later equations use their already calculated values to restrict decision variables.

The strictly positive recovery rates in (P2) preserve a direct link between each recovery input and its usable output. The corresponding output-capacity relations therefore govern the admissible recovery inflows, while disposal inflow is governed by its physical throughput capacity.

Initialization and model-generation order:

The implementation follows a fixed sequence. It first loads the raw numerical instance and performs the admissibility checks (P1)–(P7). It then calculates (P8)–(P15) and stores the resulting bounds. The customer-to-collection variables are fixed to zero for period one as part of the initialization boundary. The model equations are generated only after these steps, followed by the three payoff-table solves and the augmented epsilon-constraint grid.

Environmental input conversion:

All terms in the environmental objective are expressed in kg CO₂-eq. The electricity and water coefficients used by the case study are converted before they are loaded into the objective:

$$\kappa_c^E = \frac{EF_c^E}{3600}, \quad \kappa^W = \frac{0.84}{1000} = 8.4 \times 10^{-4}, \quad (\text{P16})$$

where EF_c^E is supplied in kg CO₂-eq/kWh, while the model records electricity in kJ and water in litres. The case-study electricity factors are 0.64 and 0.78 kg CO₂-eq/kWh for New South Wales and Victoria, respectively [1]. The water value is the reported mean climate-change impact of 0.84 kg CO₂-eq/m³ in the potable-water LCA meta-analysis [3]. These are transparent case-study inputs and should be recalibrated when a different geographic or technological context is studied.

Modeling Assumptions:

1. The supply chain is modeled as a multi-period, multi-product closed-loop network under uncertain scenarios. Strategic decisions are made before the realization of uncertainty, whereas operational flow and inventory decisions may vary across scenarios.
2. Collection centers are existing facilities with predetermined locations. Their available handling and temporary-storage capacities are assumed to be sufficient to accommodate all returned products assigned to them during the planning horizon. Therefore, their establishment and location are not treated as strategic decision variables.
3. Raw-material inventories are zero at the beginning of the planning horizon. Accordingly, the inventory balance in the first period is distinguished from those of subsequent periods. Reverse operations begin in period two. All customer-to-collection flows are fixed to zero in period one, so the collection and recovery balances also force every first-period reverse and recovered-output flow to zero.
4. For periods after the initialization period, the available return quantity is calculated by applying the predefined return rate to the demand-based proxy. All returns from the two eligible customer groups first pass through an existing collection center and are then assigned to recycling, refurbishing, remanufacturing, or disposal.
5. Product types are explicitly distinguished in all forward-flow variables and in the material-composition equations. Demand, however, is deliberately specified as an aggregate requirement for each customer group, demand point, period, and scenario rather than separately for each product type. Hence, the available product types are treated as substitutable only for satisfying this aggregate demand, while their different material compositions remain fully tracked. Once returned products reach the collection centers, their product identities are no longer tracked and the reverse flow is represented at an aggregated returned-material level.
6. The proportions of collected material allocated to recycling, refurbishing, remanufacturing, and disposal are treated as predefined parameters. These allocation proportions are bounded between zero and one and jointly account for the entire quantity of collected returned material.
7. Production costs at the iron-ingot, cast-iron, and remanufacturing facilities are defined on the basis of usable output. The corresponding unit production-cost parameters represent the direct production or conversion costs required to generate usable output and incorporate the expected conversion or yield losses. These costs are distinguished from the operating costs associated with additional handling and processing activities. Separating usable output from yield loss is consistent with established production-planning treatments of imperfect yield [2].
8. Operating costs at the refurbishing, remanufacturing, and recycling facilities are defined on the basis of usable recovered output. These costs represent the operational, handling, and processing activities associated with the corresponding recovery processes and are treated separately from direct production or conversion costs.
9. The operating cost at collection center k , o_k , represents recovery-oriented sorting, grading, consolidation, and preparation applied only to returns allocated to recycling, refurbishing, or remanufacturing. Products routed directly to disposal do not undergo these recovery-oriented activities; their receiving, basic handling, and disposal costs are incorporated in o_w . Accordingly, collection-center operating costs are calculated only on the three value-recovery routes, whereas disposal-bound flows incur o_w only.
10. Transportation costs associated with transferring returned products from customer groups to collection centers are assumed to fall outside the economic boundary of the proposed model. Consequently, these reverse-transportation costs are not included in the economic objective function.
11. All components aggregated within the environmental objective function are expressed in a common environmental-impact unit. Electricity and water consumption are converted into their corresponding carbon-equivalent impacts before being combined with process-related greenhouse-gas emissions.

12. Scenario probabilities are used only in the robust economic objective and in reported probability-weighted feasibility margins. The environmental objective deliberately uses an unnormalised sum of the complete impacts of all retained scenarios rather than their expected value. Accordingly, each scenario-dependent process-impact term is summed over n , whereas the scenario-independent whole-horizon utility impact is multiplied by $|N|$. Scenario probabilities remain nonnegative and sum to one for the economic objective.
13. Transportation-related greenhouse-gas emissions are outside the environmental boundary of the current model. Therefore, changes in the environmental objective are attributed only to the environmental components explicitly represented in the mathematical formulation.
14. The balance at the iron-ingot stage tracks only the net externally supplied or recovered iron make-up included within the model boundary, rather than the complete furnace charge. Accordingly, ϕ is an input-requirement coefficient, measured as tonnes of modeled iron feed per tonne of usable ingot output, and not a conversion-yield coefficient. The baseline value $\phi = 0.20$ represents a calibrated 20% net make-up requirement. The remaining 80% of the furnace charge is assumed to consist of pre-existing or internally recirculated iron that does not pass through the explicit supplier-to-ingot-plant arcs. Therefore, the input-usage term is correctly written as $\phi \sum_c X_{p,c,t,n}$. Division by ϕ would be appropriate only if ϕ were defined as an output yield. The value 0.20 is a case-study calibration rather than a universal industrial conversion rate and is evaluated through sensitivity analysis. Iron is modeled as one homogeneous material, with $R^1 = \{\text{Fe}\}$ and $|R^1| = 1$. The existing r^1 indices are retained for bookkeeping and do not denote multiple iron grades. Thus the requirement $\phi \sum_c X_{p,c,t,n}$ is imposed exactly once at each ingot plant, period, and scenario.
15. Three predefined IoT system alternatives are considered for each candidate facility. These alternatives are mutually exclusive, and at most one IoT system can be selected for each candidate facility. If none of the available IoT systems is selected, the corresponding candidate facility remains inactive. IoT-system selection is determined endogenously by the optimization model subject to the available investment budget and workforce-training capacity.
16. The nominal processing capacity of a candidate facility is independent of the specific IoT system selected. The IoT alternatives affect the corresponding investment, employment, and workforce-training requirements, while facility activation is represented through the selection of one of the available IoT systems.
17. The quantities M_p , M_c , $\bar{I}_{r^1,p,t}$, and $\bar{I}_{r^2,c,t}$ are finite parameter values computed and validated before model generation. The M values link inbound flow to facility activation. The inventory upper bounds represent cumulative material availability over the finite planning horizon, not independently calibrated warehouse capacities. They are used together with the required safety-stock levels and force inventory to zero when the corresponding plant is inactive. The recovery rates satisfy $0 < \delta, \omega, \gamma \leq 1$; therefore the recovery balances and output-capacity constraints provide the required activation logic for recovery facilities without additional Big- M equations. Any unrecovered fraction is treated as process loss outside the usable-output boundary.
18. For each customer group, one feasibility-violation variable is shared by all demand-point and period constraints within a scenario. It is therefore a scenario-level feasibility margin, not an additive shortage total over demand points and periods. At an optimum with a positive penalty, it represents the largest residual shortfall needed to relax the corresponding group of demand constraints. The operational flows and demand requirements remain indexed by demand point and period.
19. For periods two onward, returned quantities represent exogenous return potential generated by the previously installed product base. Because historical pre-horizon sales are unavailable, current demand is used only as a proxy for the relative size of that installed base. Returns are therefore not linked to current-period fulfilled deliveries: they originate from products placed on the market before the corresponding modeled sales period. Period one remains the initialization period with zero collection, as stated in Assumption 3.

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